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Buprenorphine-naloxone vs. extended-release naltrexone for opioid use disorder in individuals with and without criminal legal involvement: a secondary analysis of the X:BOT randomized controlled trial

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Abstract

Introduction: There is uncertainty about whether criminal legal involvement (CLI) impacts the effectiveness of medications for opioid use disorder (MOUD). We aimed to determine whether CLI modifies the association between buprenorphine-naloxone (BUP-NX) vs. extended-release naltrexone (XR-NTX) and MOUD treatment outcomes.

Methods: We conducted a secondary analysis of X:BOT, a 24-week multi-center randomized controlled trial comparing treatment outcomes between BUP-NX (n=287) and XR-NTX (n=283) in the general population. We used baseline Additional Severity-Index Lite responses to identify patients with recent CLI (n=342), defined as active CLI and/or CLI in the past 30 days, and lifetime incarceration (n=328). We explored recent CLI and lifetime incarceration as potential effect modifiers of BUP-NX vs. XR-NTX effectiveness on relapse, induction, and overdose. We conducted both intention-to-treat and per-protocol analyses for each outcome.

Results: In intention-to-treat analyses, recent CLI modified the effect of BUP-NX vs. XR-NTX on odds of successful induction (p=0.03) and hazard of overdose (p=0.04), but it did not modify

Author Statement

BAH provided supervision and project administration. DRB and BAH conceptualized the project and developed methodology. DRB performed the formal analysis and wrote the original draft. JD provided technical assistance of the formal analysis. DAF and LBP provided technical assistance on the conceptualization of the project. All authors provided review, editing, and final approval of the manuscript.

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the effect on hazard of relapse (p=0.23). All participants experienced lower odds of successful induction with XR-NTX compared to BUP-NX, but the relative likelihood of successful induction with BUP-NX was lower than XR-NTX among individuals with recent CLI (OR: 0.25, 95% CI: 0.13–0.47, p<0.001) compared to those without recent CLI (OR: 0.04, 95% CI: 0.01–0.19, p<0.001). Participants with recent CLI experienced similar hazard of overdose with XR-NTX and BUP-NX (HR: 1.12, 95% CI: 0.42–3.01, p=0.82), whereas those without recent CLI experienced greater hazard of overdose with XR-NTX compared to BUP-NX (HR: 12.60, 95% CI: 1.62–98.03, p=0.02). In per-protocol analyses, recent CLI did not modify the effect of MOUD on hazard of overdose (p=0.10) or relapse (p=0.41). Lifetime incarceration did not modify any outcome.

Conclusions: Compared to individuals without recent CLI, individuals with recent CLI experienced decreased relative effectiveness of BUP-NX compared to XR-NTX for induction and overdose outcomes. This highlights the importance of considering the impact of recent CLI on opioid use disorder treatment outcomes. Future research should explore the mechanisms through which recent CLI modifies MOUD effectiveness and aim to improve MOUD effectiveness for individuals with recent CLI.

Keywords

Addiction treatment; opioid use disorder; criminal legal involvement; incarceration

1. Introduction

In the United States, people with opioid use disorder (OUD) experience high rates of criminal legal involvement (CLI), which includes interactions with law enforcement, arrest, and incarceration. Individuals with a history of CLI have significantly higher rates of opioid-related mortality compared to the general population, especially immediately following release from incarceration (Barocas et al., 2015; Binswanger et al., 2013; Binswanger et al., 2007; Bronson et al., 2017; Bukten et al., 2017; Byron Alex, 2017; Hacker et al., 2018; Lim et al., 2012; Merrall et al., 2010; Ochoa et al., 2005; Pizzicato et al., 2018; Ranapurwala et al., 2018; Winkelman et al., 2018). Increased risk of overdose is attributable to limited social and financial support, consistent exposure to drugs, decreased drug tolerance after periods of incarceration, high prevalence of mental illness, inconsistent access to primary care and addiction treatment, and structural stigma associated with CLI (Binswanger et al., 2012; Joudrey et al., 2019). The role that this context may have on addiction treatment outcomes is not well understood. It is plausible that CLI may impact risk environments, adherence, exposure to relapse triggers, patient preferences, and other factors that impact the relative effectiveness of different OUD treatment modalities, including medications for opioid use disorder (MOUD) (Harris et al., 2013; Hillhouse et al., 2013; Riggins et al., 2017; Volkow & Collins, 2017; Wang et al., 2010).

MOUD provided to individuals during incarceration or upon release is associated with decreased rates of overdose, overdose-related fatalities, relapse, reincarceration, and emergency department visits, as well as increased rates of outpatient healthcare utilization and resumption of MOUD treatment post-release (Degenhardt et al., 2014; Green et al., 2018; Howell et al., 2021; Larney et al., 2014; Malta et al., 2019). Less research has focused on MOUD initiation in the community among individuals with CLI, the impact of past CLI on MOUD effectiveness, or the relative effectiveness of different MOUD modalities for this population. In a systematic review of MOUD-related interventions among individuals with CLI, only 8 of the 46 included studies assessed outcomes of MOUD initiation in the community after release; of the 8, only one was a randomized controlled trial (RCT) designed to compare MOUD modalities and no study compared extended-release naltrexone (XR-NTX) to buprenorphine-naloxone (BUP-NX) (Hariton & Locascio, 2018; Lobmaier et al., 2010; Malta et al., 2019). Additional studies have compared MOUD effectiveness between those with and without CLI; however, the results are both inconclusive and limited to BUP-NX (Harris et al., 2013; Hillhouse et al., 2013; Riggins et al., 2017; Wang et al., 2010).

Given this prior research, three main gaps in the literature emerge. First, there is a relative dearth of research assessing the effectiveness of MOUD initiation in the community for people with CLI. Second, there is limited and conflicting research regarding whether past CLI impacts the effectiveness of BUP-NX and XR-NTX. Third, no RCT has examined CLI as a potential effect modifier of BUP-NX and XR-NTX community-initiated treatment outcomes.

The X:BOT trial, a multi-center RCT that compared treatment outcomes with BUP-NX and XR-NTX in community dwelling individuals, assessed CLI at baseline and thus provides an opportunity to answer these questions. The trial found that participants randomized to BUP-NX experienced fewer relapse events than those receiving XR-NTX (Lee et al., 2018). The authors also reported that risk of overdose was comparable between treatment groups; however, this finding has since been disputed, as subsequent analyses of the X:BOT data have reported an increased hazard of overdose with those randomized to XR-NTX compared to BUP-NX (Ajazi et al., 2022; Dasgupta et al., 2023; Lee et al., 2022 Mar 11; Lee et al., 2023). Our objective was to investigate whether CLI modifies BUP-NX versus XR-NTX outcomes via secondary analysis of the X:BOT trial.

We hypothesize that, compared to those without CLI, individuals with CLI will experience increased relative effectiveness of BUP-NX compared to XR-NTX. Because individuals with CLI often face unique contextual circumstances, we posit that BUP-NX's lower induction barrier will result in increased treatment induction, decreased risk of relapse, and decreased risk of overdose (Joudrey et al., 2019). Given the exacerbated risk of opioid-related overdose and death among individuals with CLI during the post-release period, it is imperative to understand the comparative effectiveness of BUP-NX versus XR-NTX to inform clinical decision making and evidence-based policy recommendations (Lee et al., 2018).

2. Methods

2.1 Data and sample: X:BOT Trial

We performed a secondary analysis of data from patients enrolled in X:BOT, a 24-week open-label RCT comparing the effectiveness of BUP-NX versus XR-NTX for OUD treatment. The specifics of the conduct of this trial have been reported in depth elsewhere (Lee et al., 2018; Lee et al., 2016). Briefly, X:BOT recruited and enrolled individuals 18 or older with a DSM-5 diagnosis of OUD and recent use of non-prescribed opioids who were voluntarily admitted to inpatient opioid detoxification programs. The trial was conducted at eight community-based addiction treatment programs and employed 1:1 randomization of participants into two treatment groups: XR-NTX (monthly intramuscular injections) and BUP-NX (self-administered, daily, sublingual films). Medical and research visits were conducted weekly, during which information on urine samples, self-reported opioid use, opioid cravings, and adverse events was collected (ClinicalTrials.gov, 2020; Lee et al., 2016).

The primary outcome for X:BOT was relapse, defined as use of any non-study opioid at least once in 4 consecutive weeks or for 7 consecutive days. Secondary outcomes included successful induction onto the study medication and overdose.

The X:BOT trial was sponsored by the National Institute on Drug Abuse Center for Clinical Trials Network (CTN). CTN trials are required to upload de-identified trial data to a publicly available data share website for dissemination of research results to providers and patients (Lee et al., 2018; National Institute on Drug Abuse). This publicly available data was used for our secondary analysis. For our analysis, we included all randomized X:BOT participants.

2.2 Criminal Legal Involvement

We used responses to the Addiction Severity Index– Lite (ASI-Lite) at the baseline study visit to determine presence of recent or lifetime CLI. ASI-Lite is a validated, semi-structured interview tool that asks participants about six key areas: medical, employment/support, drug and alcohol use, legal, family/social, and psychiatric (Cacciola et al., 2007; McLellan et al., 1980). We used the following questions from the "Legal Status" section of the ASI-Lite to assess baseline CLI:

1. How many months were you incarcerated in your life?
2. Was this admission prompted or suggested by the criminal justice system?
3. Are you on probation or parole?
4. Are you presently awaiting charges, trial, or sentence?
5. How many days in the past 30 were you detained or incarcerated?
6. How many days in the past 30 have you engaged in illegal activities for profit?

To capture participants' CLI, we focused on two groups: individuals with a history of lifetime incarceration and individuals with a history of recent CLI. Lifetime incarceration was a binary variable defined as anyone who reported that they had been incarcerated for ≥ 1 month in their lifetime. We defined "recent CLI" as a binary aggregate variable based on responses to questions on criminal justice system admission, current probation/parole, presently awaiting charges/trial/sentence, incarcerated in past 30 days, and/or illegal profit in past 30 days. If participants answered "yes" to at least one of these five CLI factors, they were considered to have "recent CLI." For our primary analyses, we focused on the effect modification of lifetime incarceration and recent CLI (Figure 1, Appendix A). In secondary analyses, we assessed the effect modification of each individual CLI variable (Appendix B).

2.3 Outcomes

Our primary outcome was relapse, as defined in XBOT, and our secondary outcomes included successful medication induction and overdose.

2.3.1 Relapse—We defined relapse as X:BOT did – four consecutive weeks of any non-study opioid use by urine toxicology or self-report, or 7 consecutive days of self-reported use, as coded by a dichotomous relapse variable in the publicly available data file (Ajazi et al., 2022; Lee et al., 2018). For successfully inducted participants, time-to-relapse began at day 21 post-randomization. For unsuccessfully inducted patients, relapse was recorded as the date of induction failure, starting as early as day 0 of randomization.

2.3.2 Successful Induction—Successful induction was defined as receiving an initial dose of either XR-NTX or BUP-NX. XR-NTX induction required at least 3 days from last opioid use for detoxification, an opioid-negative urine, and a negative naloxone challenge. BUP-NX induction required significant withdrawal symptoms prior to receiving the first dose.

2.3.3 Overdose—Similar to Ajazi et al.'s method for overdose classification, we identified overdoses via data from both the Medical Dictionary for Regulatory Activities (MedDRA) and the Adverse Events report (Ajazi et al., 2022). As such, we coded adverse events that included the text "overdose" in the report of the etiology as overdoses, even if not coded as overdoses via MedDRA, by the X:BOT study team. This approach differs from the original X:BOT analysis, which only used overdose data from the MedDRA, but in light of published correspondence with Lee et al, this likely represented a misclassification and undercount of overdose events (Dasgupta et al., 2023; Lee et al., 2022 Mar 11; Lee et al., 2023).

2.4 Analysis

We conducted both intention-to-treat (ITT) and per-protocol (PP) analyses for each outcome measure. Our ITT analysis included all study participants as randomized to either the BUP-NX or XR-NTX treatment arm, whereas our PP analysis was limited to participants who were successfully inducted onto their respective study medication.

To assess for effect modification of CLI on relapse and overdose, we performed a survival analysis of time-to-event, employing a Cox proportional hazards model including our CLI variables as both covariates and interaction terms with the study arm variable. To assess for effect modification of CLI on the relationship between treatment arm and odds of successful treatment induction, we performed a logistic regression, similarly including our CLI variables as covariates and interaction terms with the study arm variable. All analyses were adjusted for opioid use severity and timing of randomization. As defined by the original X:BOT trial, opioid use severity was dichotomized into high and low severity ($\geq$6 bags or equivalent of intravenous heroin each day in the 7 days preceding admission, or <6 bags, respectively). Timing of randomization was categorized as early versus late (randomization ≤72 hours since last opioid use, or >72 hours, respectively) (Lee et al., 2018). For our primary analyses, we investigated recent CLI and lifetime incarceration as potential effect modifiers (Figure 1, Appendix A). In secondary analyses, we evaluated the effect modification of all CLI variables (Appendix B). We estimated hazards ratios with confidence intervals between the treatment arms at levels of the CLI variable using postestimation linear combination of estimates. We tested the proportional hazard assumption by testing of the interaction between log-transformed time and our variables of interest.

All analyses were conducted with StataSE 17 (StataCorp, College Station, TX). We considered p-values < 0.05 significant. Per study documentation, all X:BOT study sites obtained local Institutional Review Board approval and all participants provided written consent. As a secondary analysis of publicly available, de-identified data this study received a "Not Human Subject Research Determination" from Yale University's Institutional Review Board.

3. Results

Of the 570 X:BOT participants randomized into either the XR-NTX (n=283) or BUP-NX (n=287) treatment group, 328 (57.5%) had been incarcerated at some point in their lifetime and 342 (60%) had experienced recent CLI. There were no significant differences in the distribution of CLI variables between XR-NTX and BUP-NX treatment groups (Table 1, Appendix C). Baseline demographic differences of participants with and without lifetime incarceration and recent CLI are included in Table 2. Overall, individuals with lifetime incarceration, compared to those without lifetime incarceration, were more likely to be older, Black, male, and have histories of high severity opioid use and intravenous opioid use. Compared to those without recent CLI, individuals with recent CLI were more likely to be younger, white, and have histories of high severity opioid use, intravenous opioid use, and psychiatric disorders. In general, within the groups with and without lifetime incarceration and recent CLI, sociodemographic characteristics were evenly distributed across XR-NTX and BUP-NX treatment groups.

3.1 Time-to-Relapse

Overall, in line with original X:BOT analysis, we found that participants receiving XR-NTX were more likely to meet criteria for relapse than those receiving BUP-NX (HR: 1.37, 95% CI: 1.10–1.69, p=0.004) in the ITT analysis, whereas participants receiving XR-NTX had similar risk of relapse than those receiving BUP-NX (HR: 0.89, 95% CI: 0.69–1.15, p=0.37) in the PP analysis (Lee et al., 2018). In both the ITT and PP analyses, neither lifetime incarceration (ITT p=0.82, PP p=0.96) or recent CLI (ITT p=0.23, PP p=0.41) modified the relationship between treatment arm and time-to-relapse suggesting that relative treatment effect between the study arms was similar across groups with and without CLI. We present the results of our analyses time-to-relapse stratified by CLI categories in Figure 1, Appendix A.1

Similar to the original X:BOT analysis, we found that the proportional hazard assumption was violated (treatment arm and time interaction p=0.005) (Lee et al., 2018) for this survival analysis. Despite this violation and consistent with the analysis presented in Lee et al, we opted to not change our planned survival analysis of time-to-relapse.

3.2 Induction success

We found that the overall study population experienced a lower odds of induction success with XR-NTX than BUP-NX (OR: 0.16, 95% CI: 0.09–0.28, p<0.001), mirroring the X:BOT trial's original results (Lee et al., 2018). We found that recent CLI (p=0.03), but not lifetime incarceration (p=0.59), was a significant effect modifier suggesting that the relationship between treatment arm and induction success was different across groups with and without CLI.

In our analysis stratified by recent CLI, we found that for both participants with and without recent CLI induction success was more likely in the BUP-NX compared to XR-NTX arm but the relative likelihood of successful induction was closer to the null in participants with recent CLI (OR: 0.25, 95% CI: 0.13–0.47, p<0.001) compared to those without recent CLI (OR: 0.04, 95% CI: 0.01–0.19, p<0.001). (Figure 1, Appendix A.2)

3.3 Time-to-Overdose

Overall, in ITT analysis, individuals in the XR-NTX arm experienced a higher hazard of overdose compared to those in the BUP-NX arm (HR: 2.40, 95% CI: 1.09–5.30, p=0.03). In the PP analysis, participants experienced similar hazard of overdose with XR-NTX and BUP-NX (HR: 2.10, 95% CI: 0.86–5.14, p=0.11). These results are consistent with Ajazi et al.'s re-analysis of the public use X:BOT dataset (Ajazi et al., 2022).

In our ITT analysis, we found that recent CLI (p=0.04), but not lifetime incarceration (p=0.77), was a significant effect modifier of the relationship between XR-NTX vs. BUP-NX and time-to-overdose suggesting that the relative effect of XR-NTX vs. BUP-NX on time-to-overdose was different between those with and without recent CLI. In the PP analysis, neither recent CLI (p=0.10) nor lifetime incarceration (p=0.65) modified the association between treatment arm and time-to-overdose. We found that the proportional hazard assumption was not violated for our analysis of time-to-overdose (treatment arm and time interaction p=0.06).

In our ITT analysis stratified by recent CLI, we found that participants experienced a similar hazard of overdose with XR-NTX or BUP-NX (HR: 1.12, 95% CI: 0.42–3.01, p=0.82), with both groups experiencing 8 overdoses. Participants without a history of recent CLI experienced a greater hazard of overdose with XR-NTX compared with BUP-NX (HR:12.60, 95% CI: 1.62–98.03, p=0.02), with eleven overdoses occurring in the XR-NTX arm versus one in the BUP-NX arm. Among those in the PP sample, participants with a history of recent CLI experienced a similar hazard of overdose with XR-NTX and BUP-NX (HR: 1.12, 95% CI: 0.37–3.40, p=0.84). In contrast, among those in the PP sample without a history of CLI, individuals experienced a greater hazard of overdose with XR-NTX than BUP-NX (HR: 8.58, 95% CI: 1.02–71.78, p=0.05). Complete results of our analyses of time-to-overdose are presented in Figure 1, Appendix A.3.

Discussion

In this secondary analysis of the X:BOT trial, we found that participants with recent CLI experienced decreased relative effectiveness of BUP-NX versus XR-NTX for odds of successful induction and hazard of overdose, compared to participants without recent CLI. Importantly, among individuals with recent CLI, the risk of overdose was comparable between the study arms with the same number of overdoses observed and no difference in time-to-overdose in both the BUP-NX and XR-NTX arms of the study. This contrasted to those without recent CLI, overdose risk was lower in those in the BUP-NX arm compared to the XR-NTX arm, with only one overdose occurring in those randomized to receive BUP-NX and 11 in those randomized to receive XR-NTX arm. Generalizable inferences from these findings should be made with caution, as this study was likely underpowered to see a difference in outcomes between BUP-NX and XR-NTX in individuals with CLI, but it does highlight the potential role that contextual factors, in this case recent CLI, may have on the relative effectiveness of MOUD modalities. This finding also contradicted our hypothesis that, compared to those without CLI, individuals with CLI would experience increased relative effectiveness of BUP-NX compared to XR-NTX.

There are several possible explanations for the effect modification by recent CLI we observed in relative induction success between study arms. Successful treatment induction on XR-NTX, which was more comparable to BUP-NX induction in those with recent CLI, maybe driven by factors associated with recent CLI, such as the threat of criminal legal sanctions that incentivize sustained treatment engagement. Conversely, for individuals with recent CLI, the interplay between stigma associated with different MOUD modalities and the criminal legal system may affect treatment initiation and retention (Nunes et al., 2021). Actors in the criminal legal system (police officers, judges, parole and probation officers) may hold stigmatizing beliefs about the efficacy or appropriateness of opioid agonists, like buprenorphine, for the treatment of OUD and prefer the use of naltrexone for the treatment of OUD (Andraka-Christou & Atkins, 2020; Andraka-Christou et al., 2019; Finlay et al., 2020). When individuals with CLI interact with these criminal legal system actors, who have discretionary power over their rights and freedoms, it may affect an individual’s likelihood of remaining on XR-NTX or BUP-NX.

Possible explanations of the observed effect modification by recent CLI of relative hazards of overdose between XR-NTX or BUP-NX are also worth exploring. First, the observed effect modification in relative induction success could partially explain the subsequent effect modification observed in relative hazards of overdose. If individuals with recent CLI are relatively more likely to successfully initiate XR-NTX compared to those without recent CLI, it follows that they are relatively more likely benefit from XR-NTX and to have a comparable benefit to those in the BUP-NX arm. That said, in the PP analysis which removed participants who did not receive the study medication (Appendix A.3) the pattern of comparable risk of overdose between study arms in those with recent CLI but reduced risk in the BUP-NX arm relative to XR-NTX arm in those without recent CLI persisted, although the measure of effect modification was not statistically significant. Importantly, given the high induction barrier for XR-NTX, participants who receive the first injection likely have other attributes that impact subsequent OUD outcomes making interpretation of a PP analysis, where participants in different treatment arms are no longer reflect unbiased samples, challenging.

In addition to induction success, treatment adherence is also often associated with risk of overdose and most overdoses occurred in X:BOT, even in those with successful induction, after treatment was discontinued (Lee et al., 2023). As with induction, there are likely other attributes of individuals associated with both treatment adherence and risk of overdose, making parsing whether a study outcome can be attributed to any factor in-and-of-itself, such as adherence, impossible. In addition, as noted by the original study team, X:BOT was a “randomized comparison, not strictly of medications, but rather of assignment to a regimen that includes initiation, medication treatment and adherence,” (Lee et al., 2022) a situation that mirrors clinical decisions for patients with OUD open to treatment with either XR-NTX or BUP-NX which occur prior to knowledge of whether a patient will be able to successfully start or adhere to a medication.

Second, our findings also suggest a potential impact of differential delivery systems for BUP-NX vs. XR-NTX on overdose. While differences in overdose risk between BUP-NX and XR-NTX are often attributed to the differences in mechanism of action at the mu-opioid receptor (i.e., partial antagonism versus antagonism), it is also plausible that differences in treatment outcomes we observed in those with CLI relative those without CLI are due to differences in the delivery system. BUP-NX is a daily, sublingual medication and XR-NTX is a once-monthly injection and for individuals with recent CLI, who often experience housing and employment instability, decreased healthcare access, and increased healthcare skepticism, which can disrupt treatment engagement, a once-monthly injection may lower treatment barriers and subsequently improve outcomes. Ongoing, current RCTs comparing extended-release buprenorphine (XR-BUP) formulations to sublingual BUP-NX and XR-NTX will shed additional light on the extent to which mode of MOUD administration (daily dose vs. monthly injection), as opposed to mechanism of action (buprenorphine vs. naltrexone), impacts treatment outcomes (Gordon et al., 2021).

Our results underscore the importance of considering the role of individual contextual factors, beyond attributes of the medication itself, and their potential impact on the effectiveness of modalities for a range of OUD outcomes. As noted above, for individuals with recent CLI, stigma associated with different MOUD modalities held by actors in the criminal legal system may play a role in treatment initiation and adherence. Underlying criminal-legal stigma may also exacerbate the impact of MOUD-related stigma and negatively impact perceptions of BUP-NX among individuals with recent CLI, as well as their families, support systems, and employers (Howell et al., 2022; Martin et al., 2020). The role that stigma may play in the comparative effectiveness of BUP-NX compared to XR-NTX warrants additional exploration and underscores the importance of advocating for non-stigmatizing, harm reduction-based healthcare provision and societal education. Our findings may also reflect the impact of patient-level factors, which may differ in people with CLI, such as patient preferences and systems of beliefs surrounding MOUD, as previous research has demonstrated that these factors can drive treatment outcomes (Kaplowitz et al., 2022; Nunes et al., 2021; Puglisi et al., 2019).

Although recent CLI modified odds of induction success and hazard of overdose, we found that lifetime incarceration did not modify any study outcome, highlighting the importance of the CLI time-course, as the impact of exposure to the criminal legal system on these specific outcomes may recede with time. This finding aligns with previous research, which found that recent criminal charges impacted BUP-NX treatment outcomes, whereas distant charges did not (Harris et al., 2013).

Furthermore, it is important to note that despite demonstrating an effect modification on induction success and overdose, recent CLI did not modify the relative hazard of study-defined relapse between the treatment arms. One would expect effects on relapse to travel in tandem with overdose given the intrinsically intertwined relationship between relapse, opioid use, and overdose risk. The lack of effect modification of recent CLI on risk of relapse, in contrast to its effect on overdose risk, may be explained by the definition of relapse as “4 consecutive weeks of any non-study opioid use by urine toxicology or self-report, or 7 consecutive days of self-reported use” (Lee et al., 2018). This study definition of relapse may therefore reflect a longer-term return to use, whereas overdose can occur in the setting of acute use. In addition, the contextual factors of CLI as described above, may change the expected relationship between relapse, opioid use, and overdose, with any opioid use in people with CLI being associated with higher relative risk of overdose. This is important to highlight, as relapse is a common outcome used to assess MOUD effectiveness, but it may not predict overdose outcomes in all subpopulations equally (Biondi et al., 2020).

While this study helped illuminate recent CLI as an important effect modifier in the relationship between BUP-NX vs. XR-NTX and OUD treatment outcomes, there are several limitations that must be considered. First, we were limited by the X:BOT trial’s data collection and our reliance on the ASI-Lite to capture CLI. Although we created a “Recent CLI” variable to encompass legal indicators of recent involvement, such as incarceration in the past 30 days and awaiting charges/trial/sentencing, we are unable to precisely define the timeline of “recent.” While previous studies have defined “recent” as involving the previous 2 years, further research is needed to understand the time period in which “recent CLI” remains a significant effect modifier of MOUD effectiveness (Harris et al., 2013). Furthermore, the ASI-Lite is a self-report survey and is prone to reporting bias. Given the stigmatizing nature of CLI, our results may underestimate the presence and impact of CLI. Additionally, our study was also limited by X:BOT’s patient population, which comprised individuals who were voluntarily admitted to inpatient opioid detoxification programs. This precludes generalization of our findings to populations who access OUD treatment via different means including court-mandated treatment, an important scenario to consider for individuals with a history of CLI. It is also important to note the limitations posed by X:BOT’s definition of relapse, as our study results may have varied if we used a different operational definition (Biondi et al., 2020). Lastly, stratification of XR-NTX and BUP-NX treatment arms by CLI variables decreased sample sizes in each group. Therefore, it is possible that we lacked power to detect significant differences between treatment groups for some outcomes.

Conclusion

Individuals with recent CLI experienced decreased relative effectiveness of BUP-NX vs. XR-NTX for induction and overdose outcomes, compared to those without recent CLI. However, neither recent CLI nor lifetime incarceration modified the relationship between either medication or hazard of relapse. These findings demonstrate the importance of considering the impact of recent CLI on MOUD outcomes. Future research should investigate how and why recent CLI modifies MOUD effectiveness to enable evidence-based interventions aimed at improving BUP-NX effectiveness for individuals with recent CLI. Given the high percentage of people with OUD who interact with the criminal legal system each year and the high risk of overdose in this population, it is imperative that future studies of MOUD effectiveness consider the impact of CLI system on treatment outcomes.

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Appendix A

Appendix A.1:

Relapse, by CLI

	Lifetime Incarceration			No Lifetime Incarceration			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	110 (67)	93 (57)	1.40 (1.06, 1.85), p=0.018	71 (60)	68 (55)	1.33 (0.95, 1.87), p=0.09	0.05 (−0.39, 0.49), p=0.82
PP	63 (53)	87 (56)	0.90 (0.65, 1.25), p=0.53	39 (45)	60 (52)	0.89 (0.59, 1.33), p=0.57	0.01 (−0.51, 0.53), p=0.96
	Recent CLI			**No Recent CLI**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	105 (64)	106 (60)	1.24 (0.95, 1.63), p=0.12	75 (64)	54 (50)	1.63 (1.14, 2.32), p=0.007	−0.27 (−0.72, 0.18), p=0.23
PP	61 (51)	93 (57)	0.83 (0.60, 1.15), p=0.26	41 (49)	53 (50)	1.03 (0.69, 1.56), p=0.88	−0.22 (−0.74, 0.31), p=0.41

Appendix A.2:

Induction Success, by CLI

	Lifetime Incarceration			No Lifetime Incarceration			
	XR-NTX n (%)	BUP-NX n (%)	OR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	OR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
ITT	118 (72)	154 (94)	0.14 (0.06, 0.29), p<0.001	86 (73)	116 (94)	0.19 (0.08, 0.43), p<0.001	−0.31 (−1.44, 0.81), p=0.59
	Recent CLI			**No Recent CLI**			
	XR-NTX n (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	120 (73)	163 (92)	0.25 (0.13, 0.47), p<0.001	83 (71)	106 (98)	0.04 (0.01, 0.19), p<0.001	1.74 (0.15, 3.33), p=0.03

Appendix A.3:

Overdose, by CLI

	Lifetime Incarceration			No Lifetime Incarceration			
	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
ITT	11 (7)	4 (2)	2.77 (0.87, 8.75), p=0.08	8 (7)	5 (4)	2.18 (0.71, 6.72), p=0.17	0.24 (−1.38, 1.86), p=0.77
PP	8 (7)	4 (3)	2.47 (0.73, 8.33), p=0.15	4 (5)	4 (3)	1.60 (0.40, 6.47), p=0.51	0.43 (−1.43, 2.30), p=0.65
	Recent CLI			**No Recent CLI**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	8 (5)	8 (4)	1.12 (0.42, 3.01), p=0.82	11 (9)	1 (1)	12.60 (1.62, 98.03), p=0.02	−2.42 (−4.71, −0.13), p=0.04
PP	6 (5)	7 (4)	1.12 (0.37, 3.40), p=0.84	6 (7)	1 (1)	8.58 (1.02, 71.78), p=0.05	−2.04 (−4.44, 0.37), p=0.10

Appendix B

Appendix B.1:

Relapse, by CLI

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	7 (70)	2 (22)	3.80 (0.76, 18.85), p=0.10	174 (64)	159 (57)	1.34 (1.08, 1.67), p=0.008	1.04 (−0.58, 2.66), p=0.21
PP	6 (67)	0 (0)	1.70e9 (1.31e9, 2.20e9), p<0.001	96 (49)	147 (56)	0.86 (0.66, 1.11), p=0.24	21.41 (n/a), p=n/a

	Current probation/parole			No current probation/parole			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	30 (71)	32 (64)	1.27 (0.76, 2.11), p=0.36	150 (63)	129 (54)	1.40 (1.10, 1.77), p=0.006	−0.10 (−0.66, 0.46), p=0.73
PP	21 (64)	23 (56)	1.19 (0.65, 2.20), p=0.57	80 (47)	124 (54)	0.84 (0.63, 1.11), p=0.22	0.35 (−0.32, 1.02), p=0.30
	Awaiting charges/trial/sentencing			**Not awaiting charges/trial/sentencing**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	21 (62)	27 (64)	1.01 (0.57, 1.79), p=0.98	160 (64)	134 (55)	1.45 (1.15, 1.83), p=0.002	−0.36 (−0.98, 0.25), p=0.25
PP	11 (46)	22 (61)	0.68 (0.33, 1.40), p=0.29	91 (51)	125 (53)	0.94 (0.71, 1.23), p=0.65	−0.33 (−1.10, 0.45), p=0.41
	Incarcerated in past 30d			**Not incarcerated in past 30d**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	16 (80)	13 (65)	1.48 (0.71, 3.08), p=0.30	165 (63)	148 (55)	1.37 (1.09, 1.71), p=0.006	0.08 (−0.69, 0.85), p=0.84
PP	9 (69)	12 (63)	1.07 (0.45, 2.55), p=0.87	93 (49)	135 (54)	0.89 (0.68, 1.16), p=0.39	0.19 (−0.72, 1.09), p=0.69
	Illegal activities for profit in past30d			**No illegal activities for profit in past 30d**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	74 (60)	74 (59)	1.17 (0.84, 1.61), p=0.35	105 (67)	86 (53)	1.55 (1.16, 2.06), p=0.003	−0.28 (−0.72, 0.15), p=0.20
PP	43 (47)	66 (57)	0.77 (0.52, 1.13), p=0.18	58 (52)	80 (52)	1.01 (0.72, 1.42), p=0.95	−0.28 (−0.79, 0.24), p=0.29
	Convicted in lifetime			**Not convicted in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	149 (63)	138 (56)	1.35 (1.07, 1.70), p=0.012	32 (70)	23 (56)	1.51 (0.88, 2.59), p=0.14	−0.11 (−0.70, 0.48), p=0.72
PP	83 (49)	125 (54)	0.86 (0.65, 1.14), p=0.30	19 (58)	22 (55)	1.10 (0.59, 2.03), p=0.77	−0.24 (−0.92, 0.44), p=0.49
	Arrested and charged in lifetime			**Not arrested and charged in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX(n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	147 (67)	128 (58)	1.38 (1.08, 1.75), p=0.009	33 (53)	32 (49)	1.35 (0.82, 2.22), p=0.23	0.02 (−0.53, 0.57), p=0.95
PP	83 (53)	116 (56)	0.91 (0.68, 1.20), p=0.50	18 (38)	30 (48)	0.83 (0.46, 1.50), p=0.54	0.08 (−0.57, 0.74), p=0.80
	Drug charges in lifetime			**No drug charges in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	93 (65)	79 (56)	1.40 (1.04, 1.90), p=0.03	88 (63)	82 (57)	1.34 (0.99, 1.82), p=0.058	0.04 (−0.38, 0.47), p=0.84
PP	54 (51)	70 (53)	0.97 (0.68, 1.39), p=0.89	48 (48)	77 (55)	0.83 (0.57, 1.19), p=0.31	0.16 (−0.35, 0.67), p=0.53

Appendix B.2:

Induction Success, by CLI

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTXn (%)	BUP-NX n (%)	OR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	OR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
ITT	9 (90)	7 (78)	2.14 (0.16, 29.22), p=0.57	195 (71)	263 (95)	0.14 (0.08, 0.25), p<0.001	2.73 (0.05, 5.41), p=0.045
	Current probation/parole			**No current probation/parole**			
	XR-NTXn (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	33 (79)	41 (82)	0.80 (0.28, 2.28), p=0.68	170 (71)	229 (97)	0.08 (0.04, 0.17), p<0.001	2.29 (0.99, 3.58), p=0.001
	Awaiting charges/trial/sentencing			**Not awaiting charges/trial/sentencing**			
	XR-NTXn (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	24 (71)	36 (86)	0.45 (0.14, 1.41), p=0.17	180 (72)	234 (96)	0.12 (0.06, 0.23), p<0.001	1.35 (0.01, 2.68), p=0.048
	Incarcerated in past 30d			**Not incarcerated in past 30d**			
	XR-NTXn (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX< n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	13 (65)	19 (95)	0.12 (0.01, 1.15), p=0.07	190 (73)	251 (94)	0.16 (0.09, 0.29), p<0.001	−0.26 (−2.56, 2.05), p=0.83
	Illegal activities for profit in past 30d			**No illegal activities for profit in past 30d**			
	XR-NTXn (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	92 (75)	116 (93)	0.23 (0.10, 0.50), p<0.001	111 (70)	153 (95)	0.12 (0.05, 0.26), p<0.001	0.65 (−0.47, 1.77), p=0.26
	Convicted in lifetime			**Not convicted in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTXn (%)	BUP-NX n (%)	OR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	OR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
ITT	171 (72)	230 (94)	0.17 (0.10, 0.31), p<0.001	33 (72)	40 (98)	0.07 (0.01, 0.56), p=0.01	0.92 (−1.26, 3.09), p=0.41
	Arrested and charged in lifetime			**Not arrested and charged in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	156 (71)	206 (93)	0.17 (0.09, 0.31), p<0.001	47 (76)	63 (97)	0.10 (0.02, 0.45), p=0.003	0.57 (−1.07, 2.22), p=0.50
	Drug charges in lifetime			**No drug charges in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**OR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**OR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	105 (73)	131 (92)	0.22 (0.11, 0.45), p<0.001	99 (71)	139 (96)	0.10 (0.04, 0.26), p<0.001	0.74 (−0.42, 1.89), p=0.21

Appendix B.3:

Overdose, by CLI

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
ITT	0 (0)	0 (0)	0.85 (0, n/a), p=1.00	19 (7)	9 (3)	2.39 (1.08, 5.28), p=0.03	−1.04 (−7.55e7, 7.55e7), p=1.00
PP	0 (0)	0 (0)	0.86 (0.35, 2.12), p=0.75	12 (6)	8 (3)	2.05 (0.84, 5.05), p=0.12	−0.87 (n/a, n/a), p=n/a
	Current probation/parole			**No current probation/parole**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	4 (10)	3 (6)	2.16 (0.48, 9.69), p=0.32	15 (6)	6 (3)	2.67 (1.04, 6.89), p=0.04	−0.21 (−1.99, 1.56), p=0.81
PP	3 (9)	2 (5)	2.60 (0.43, 15.62), p=0.30	9 (5)	6 (3)	1.99 (0.70, 5.61), p=0.20	0.27 (−1.80, 2.34), p=0.80
	Awaiting charges/trial/sentencing			**Not awaiting charges/trial/sentencing**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β**

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
							coefficient (95% CI), p-value
ITT	2 (6)	2 (5)	1.37 (0.19, 9.79), p=0.75	17 (7)	7 (3)	2.67 (1.11, 6.45), p=0.03	−0.67 (−2.83, 1.49), p=0.54
PP	1 (4)	2 (6)	0.84 (0.08, 9.33), p=0.89	11 (6)	6 (3)	2.43 (0.89, 6.63), p=0.08	−1.06 (−3.67, 1.55), p=0.43
	Incarcerated in past 30d			**Not incarcerated in past 30d**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	1 (5)	3 (15)	0.29 (0.03, 3.19), p=0.31	18 (7)	6 (2)	3.47 (1.38, 8.77), p=0.01	−2.50 (−5.10, 0.10), p=0.06
PP	0 (0)	3 (16)	9.49e-21 (3.3e-21, 2.72e-20), p<0.001	12 (6)	5 (2)	3.47 (1.21, 9.94), p=0.02	−47.35 (n/a), p=n/a
	Illegal activities for profit in past 30d			**No illegal activities for profit in past 30d**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	6 (5)	6 (5)	1.06 (0.34, 3.30), p=0.91	13 (8)	3 (2)	5.25 (1.49, 18.44), p=0.01	−1.60 (−3.29, 0.10), p=0.06
PP	5 (5)	6 (5)	1.06 (0.32, 3.48), p=0.93	7 (6)	2 (1)	5.10 (1.06, 24.65), p=0.04	−1.57 (−3.55, 0.40), p=0.12
	Convicted in lifetime			**Not convicted in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	17 (7)	7 (3)	2.75 (1.14, 6.64), p=0.02	2 (4)	2 (5)	1.13 (0.16, 8.11), p=0.91	0.89 (−1.27, 3.06), p=0.42
PP	11 (6)	6 (3)	2.53 (0.93, 6.91), p=0.07	1 (3)	2 (5)	0.63 (0.06, 7.18), p=0.71	1.39 (−1.25, 4.02), p=0.30
	Arrested and charged in lifetime			**Not arrested and charged in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX C n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**
ITT	16 (7)	8 (4)	2.32 (0.99, 5.42), p=0.052	3 (5)	1 (2)	2.99 (0.31, 28.80), p=0.34	−0.26 (−2.67, 2.16), p=0.84
PP	11 (7)	7 (3)	2.19 (0.85, 5.70), p=0.11	1 (2)	1 (2)	1.20 (0.07, 19.18), p=0.90	0.61 (−2.33, 3.54), p=0.69
	Drug charges in lifetime			**No drug charges in lifetime**			
	XR-NTX n (%)	**BUP-NX n (%)**	**HR (95% CI), p-value**	**XR-NTX n (%)**	**BUP-NX n (%)**	**HR (95% CI), p-value**	**Effect Modification: β coefficient (95% CI), p-value**

	Admission prompted by CJS			Admission not prompted by CJS			
	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	XR-NTX n (%)	BUP-NX n (%)	HR (95% CI), p-value	Effect Modification: β coefficient (95% CI), p-value
ITT	10 (7)	5 (4)	2.16 (0.73, 6.33), p=0.16	9 (6)	4 (3)	2.71 (0.83, 8.81), p=0.098	−0.23 (−1.83, 1.37), p=0.78
PP	8 (8)	4 (3)	2.52 (0.75, 8.46), p=0.14	4 (4)	4 (3)	1.51 (0.38, 6.08), p=0.56	0.51 (−1.34, 2.36), p=0.59

Appendix C: CLI Characteristics

	ITT (n=570)			PP (n=474)		
	XR-NTX (n=283)	BUP-NX (n=287)	P-value	XR-NTX (n=204)	BUP-NX (n=270)	P-value
Admission prompted by criminal justice system	10 (52.6)	9 (47.4)	0.79	9 (56.3)	7 (43.8)	0.28
Admission not prompted by criminal justice system	273 (49.6)	278 (50.5)		195 (42.6)	263 (57.4)	
Current probation/parole	42 (45.7)	50 (54.4)	0.41	33 (44.6)	41 (55.4)	0.75
No current probation/ parole	240 (50.3)	237 (49.7)		170 (42.6)	229 (57.4)	
Awaiting charges/trial/ sentencing	34 (44.7)	42 (55.3)	0.36	24 (40.0)	36 (60.0)	0.61
Not awaiting charges/trial/sentencing	249 (50.4)	245 (49.6)		180 (43.5)	234 (56.5)	
Incarcerated in past 30d	20 (50.0)	20 (50.0)	0.95	13 (40.6)	19 (59.4)	0.79
Not incarcerated in past 30d	262 (49.5)	267 (50.5)		190 (43.1)	251 (56.9)	
Illegal activities for profit in past 30d	123 (49.6)	125 (50.4)	0.99	92 (44.2)	116 (55.8)	0.63
No illegal activities for profit in past 30d	158 (49.5)	161 (50.5)		111 (42.1)	153 (58.0)	
Convicted in lifetime	237 (49.1)	246 (50.9)	0.51	171 (42.6)	230 (57.4)	0.68
Not convicted in lifetime	46 (52.9)	41 (47.1)		33 (45.2)	40 (54.8)	
Arrested and charged in lifetime	220 (49.9)	221 (50.1)	0.83	156 (43.1)	206 (56.9)	0.95
Not arrested and charged in lifetime	62 (48.8)	65 (51.2)		47 (42.7)	63 (57.3)	
Drug charges in lifetime	144 (50.4)	142 (49.7)	0.74	105 (44.5)	131 (55.5)	0.52
No drug charges in lifetime	139 (48.9)	145 (51.1)		99 (41.6)	139 (58.4)	

Abbreviations:

OUD	opioid use disorder
CLI	criminal legal involvement
MOUD	medications for opioid use disorder

RCT	randomized controlled trial
XR-NTX	extended-release naltrexone
BUP-NX	buprenorphine-naloxone

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Highlights

- Criminal legal context impacts medication for opioid use disorder outcomes
- Recent criminal legal involvement decreased relative BUP-NX vs XR-NTX effectiveness
- Recent criminal legal involvement modified induction and overdose, but not relapse

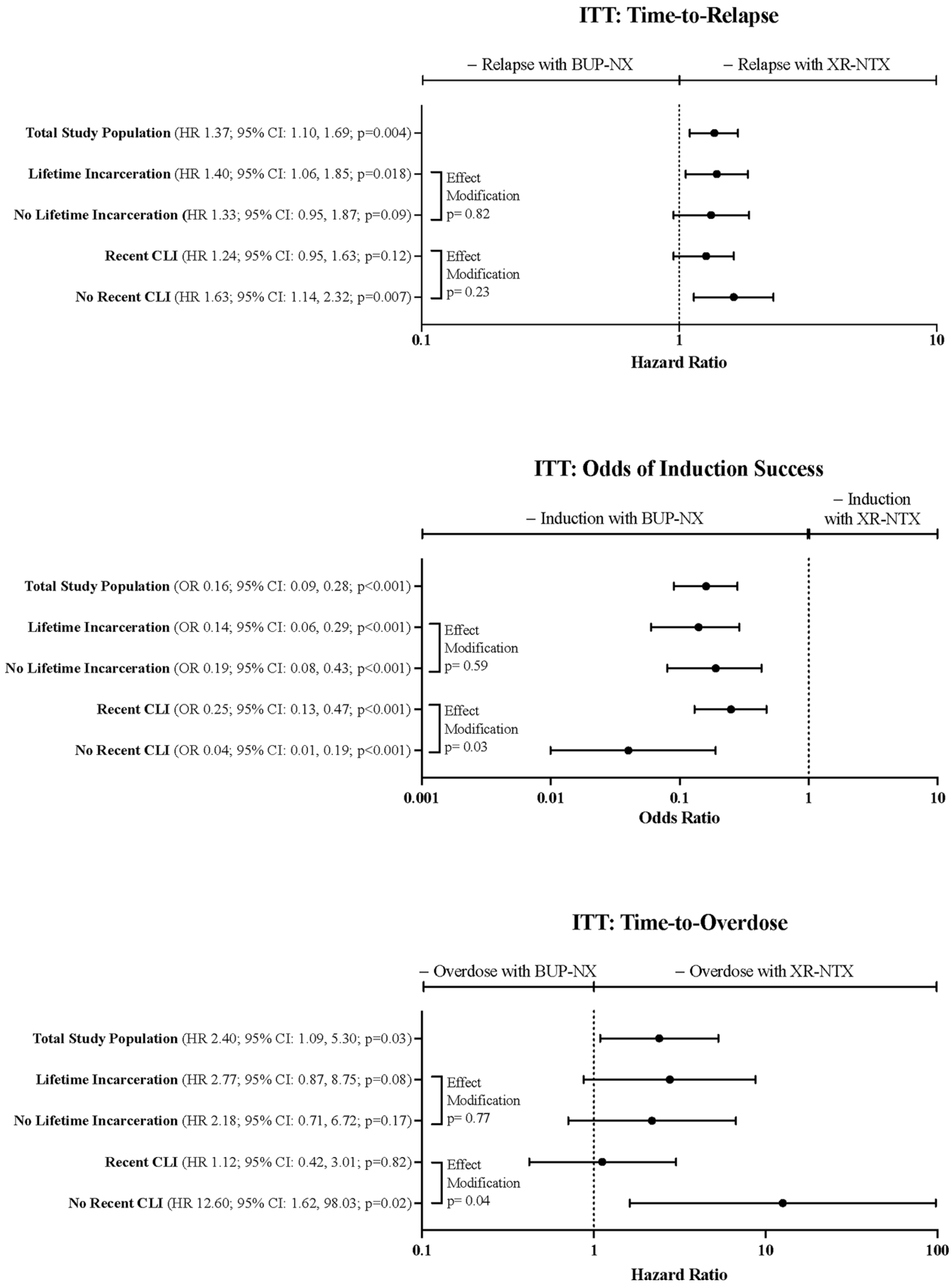

Figure 1.

Results of intention-to-treat analysis of XBOT study on time-to-relapse, odds of induction success, and time-to-overdose stratified by lifetime incarceration and recent criminal legal involvement.

Table 1.

CLI characteristics

	ITT (n=570)			PP (n=474)		
	XR-NTX (n=283)	**BUP-NX (n=287)**	**P-value**	**XR-NTX (n=204)**	**BUP-NX (n=270)**	**P-value**
Lifetime incarceration	165 (58%)	163 (57%)	0.72	118 (58%)	154 (57%)	0.86
No lifetime incarceration	118 (42%)	124 (43%)		86 (42%)	116 (43%)	
Recent CLI	164 (60%)	178 (62%)	0.35	120 (59%)	163 (60%)	0.75
No Recent CLI	117 (41%)	108 (38%)		83 (41%)	106 (39%)	

Table 2.

Demographic and substance use characteristics

	Total study population (n=570)	Lifetime Incarceration			Recent CLI		
		Yes (n=328)	No (n=242)	P-value	Yes (n=342)	No (n=225)	P-value
Sex, n (%)							
Female	169 (30)	78 (24)	91 (38)	<0.001	98 (29)	71 (32)	0.46
Male	401 (70)	250 (76)	151 (62)		244 (72)	154 (68)	
Age, mean (SE)	33.9 (0.4)	35.4 (0.5)	31.7 (0.6)	<0.001	32.4 (0.5)	36.0 (0.7)	<0.001
Ethnicity, n (%)							
Hispanic	99 (17.4)	52 (15.9)	47 (19.4)	0.27	64 (18.7)	34 (15.1)	0.27
Race, n (%)							
Black	73 (14.3)	52 (17.7)	21 (9.7)	0.01	35 (11.3)	38 (19.1)	0.02
White	437 (85.7)	242 (82.3)	195 (90.3)		274 (88.7)	161 (80.9)	
Employment, n (%)							
Working now	105 (18.4)	47 (14.3)	58 (24.0)	0.077	56 (16.4)	47 (20.9)	0.29
Unemployed	360 (63.2)	224 (68.3)	136 (56.2)		231 (67.5)	128 (56.9)	
Opioid use severity, n (%)							
High severity	227 (39.8)	149 (45.4)	78 (32.2)	0.001	153 (44.7)	73 (32.4)	0.003
Low severity	343 (60.2)	179 (54.6)	164 (67.8)		189 (55.3)	152 (67.6)	
IV drug use, n (%)	360 (63.4)	228 (69.9)	132 (54.6)	<0.001	241 (70.9)	118 (52.4)	<0.001
Primary opioid used in 7d prior to admission, n (%)							
Buprenorphine	8 (1.4)	3 (0.9)	5 (2.1)	0.043	5 (1.5)	2 (0.9)	<0.001
Opioid analgesics	90 (15.9)	41 (12.6)	49 (20.3)		35 (10.3)	54 (24.0)	
Methadone	7 (1.2)	5 (1.5)	2 (0.8)		4 (1.2)	3 (1.3)	
Heroin	463 (81.5)	277 (85.0)	186 (76.9)		296 (87.1)	166 (73.8)	
Cost/d of primary opioid, mean (SE)	93.6 (3.2)	92.8 (4.0)	94.5 (5.2)	0.79	96.4 (3.8)	89.5 (5.6)	0.29
Hamilton Depression Scale, (0–50), mean (SE)	8.9 (0.3)	8.3 (0.3)	9.8 (0.4)	0.008	8.9 (0.3)	9.0 (0.4)	0.95
Subjective opioid withdrawal (0–64), mean (SE)	15.6 (0.6)	14.2 (0.7)	17.4 (0.9)	0.004	16.1 (0.7)	14.7 (0.9)	0.22
History of psychiatric disorders, self-report, n (%)	390 (68.4)	221 (67.4)	169 (69.8)	0.53	245 (71.6)	142 (63.1)	0.03